WG 6

Intelligent Systems

CLASSIFICATION ERRORS

DEFINITIONS

CLASSIFICATION ERRORS

When a model incorrectly predicts the category or class of a data point.

PRECISION

Metric that evaluates the proportion of true positive predictions among all positive predictions made by a model.

TP/(TP+FP)

RECALL

Metric that evaluates the proportion of actual positives correctly identified by the model.

TP/(TP+FN)

PRECISION VS RECALL… STILL NOT CLEAR?

If you're looking for precision score, you should try answer the following question:

"Of all the instances the model labeled as positive, how many are actually positive?"

On the other hand, when looking for recall score, you should try answer the following one:

"Of all the actual positive instances, how many did the model successfully identify?"

Exercise 2: Classification Errors

Given the following confusion matrix (system A):

	actual +	actual -
predicted +	200 TP	25 FP
predicted -	10 FN	150 TN

Q1: What is the Precision and Recall for this classification?

PRECISION = TP / (TP + FP) = 200 / (200+25) = 0.89

RECALL = TP / (TP + FN) = 200 / (200 +10) = 0.95

System B got the following scores:

	actual +	actual -
predicted +	207 TP	50 FP
predicted -	3 FN	125 TN

Now, imagine the task was to predict whether a patient might possibly have cancer and after a positive test (it looks like the person has cancer), the person would be further investigated.

Q2: Which of these two systems is better in this case?

PRECISION = TP / (TP + FP) = 207 / (207+50) = 0.80

RECALL = TP / (TP + FN) = 207 / (207+3) = 0.98

WHICH MODEL IS BETTER?

In this case, we trade precision for recall . WHY?

Precision A > Precision B → With model B, there is a higher chance that someone that is actually negative gets predicted as positive to cancer. But, remember that in case of positive test, further investigations are done (so these subjects will be correctly classified as negative later on…)

Recall A < Recall B → With model A, there is a higher probability that someone with cancer will be classified as negative. That would be a problem because we know that a person would be further investigated after a positive test outcome.

For this reason, we choose model B over model A, as we want to minimize the False Negatives.

REGRESSION ERRORS

DEFINITIONS

REGRESSION MODEL

Model that predicts numerical values based on input data.

REGRESSION ERROR

Discrepancy between actual values and predicted values.

$$\sum_{i=1}^{n} (f(x_i) - y_i)^2$$

Exercise 3: Regression Errors

Consider a regression function f(x) = x that was trained on some training set, and that is now evaluated with the following test-set of datapoints { (0,1), (1,2), (2,2.5),(3,2),(0,0), (1,1), (2,1),(3,3)}.

3a) What is the error value for this learned function?

$$\sum_{i=1}^{n} (f(x_i) - y_i)^2$$

Now we have error(p)= $(0-1)^2 + (1-2)^2 + (2-2.5)^2 + (3-2)^2 + 0 + (1-1)^2 + (2-1)^2 + (3-3)^2 = 1+1+0.25+1+0+0+1+0 = 4.25$

3b) Can you think of a better function?

Finding a better regression function typically involves exploring different models and adjusting their parameters to minimize the error. In a simple linear regression context, like the one we're considering with functions of the form $f(x)=mx+b$, we would adjust the slope (m) and y-intercept (b) to find the best fit.

{ (0,1), (1,2), (2,2.5),(3,2),(0,0), (1,1), (2,1),(3,3)}

Try to draw a line that looks better. Once you found one, check it. What about $f(x)= \frac{1}{2} x + 1$? We have $f(0)= 1$ $f(1)=1.5$, $f(2)=2$ and $f(3) = 2.5$. This gives the following errors for each of the points $\text{error}(p)= (1 -1)^2 + (1.5-2)^2+ (2-2.5)^2 + (2.5-2)^2+ 1+(1.5-1)^2+ (2-1)^2+(2.5-3)^2$ which is $0-0.25+0.25+0.25+ 0+0.25+1+0.25 = 3.25$. Better but still not good.

RANKING ERRORS

DEFINITIONS

RANKING MODEL

Model that sorts items or data points in a specific order based on relevance relative to a given query or set of criteria.

RANKING ERROR

Discrepancy between predicted order of items and the true, expected order based on relevance or importance.

RANKING PRECISION

Ratio of relevant items found up to that rank over number of items up to that rank. .

Consider the following output of an algorithm written to retrieve Plays by Shakespeare:

1. Harry Potter and the Prisoner of Azkaban
2. A Midsummer Night's Dream
3. King Lear
4. Umbrella Academy
5. Macbeth
6. Much Ado About Nothing
7. Romeo and Julia
8. The Tempest
9. The Spring
10. Artificial Intelligence, a Modern Approach

1, 4, 9 and 10 are definitely NOT plays by Shakespeare!

What is the precision at 5, 7 and 10 of this ranking?

PRECISION AT 5: (5 -2)/5 = 3/5

PRECISION AT 7: (7-2)/7 = 5/7

PRECISION AT 10: (10-4)/10 = 6/10

BAYESIAN LEARNING FOR TEXT CLASSIFICATION

Suppose there are 5 documents about two courses “Intelligent Systems (IS)“ and
“Distributed Systems (DS)”:
D1: Logics and intelligent systems are very important in the course
D2: Intelligent algorithms are critical to solve problems in distributed environments
D3: Intelligent algorithms and logic are crucial
D4: Distributed systems are Intelligent
D5: Distributed systems need good algorithms

D1, D3 and D4 are about the Intelligent Systems class, D2 and D5 about Distributed Systems.
Use the (Boolean) features Logic, Intelligent, Systems, Distributed and Algorithms to predict whether the document D6 “Distributed algorithms can help reasoning in logic” is about the IS class, or DS.

Do this by making a naive Bayes classifier.[1]

DEFINITION

NAIVE BAYES CLASSIFIER

Simple probabilistic classifier that applies Bayes’ Theorem with the assumption of independence between every pair of feature.

The texts and features can be represented in the following tabular form:

	L	I	S	D	A	Course
D1	T	T	T	F	F	IS
D2	F	T	F	T	T	DS
D3	T	T	F	F	T	IS
D4	F	T	T	T	F	IS
D5	F	F	T	T	T	DS
D6	*T*	*F*	*F*	*T*	*T*	*?*

WE NEED TO CALCULATE:

- P(IS | L, -I, -S, D, A) = P(IS) * P(L|IS) * P(-I|IS) * P(-S|IS) * P(D|IS) * P(A|IS)
- P(DS | L, -I, -S, D, A) = P(DS) * P(L|DS) * P(-I|DS) * P(-S|DS) * P(D|DS) * P(A|DS)

P(IS | L, -I, -S, D, A) = P(IS) * P(L|IS) * P(-I|IS) * P(-S|IS) * P(D|IS) * P(A|IS)

	L	I	S	D	A	Course
D1	T	T	T	F	F	IS
D2	F	T	F	T	T	DS
D3	T	T	F	F	T	IS
D4	F	T	T	T	F	IS
D5	F	F	T	T	T	DS
D6	*T*	*F*	*F*	*T*	*T*	*?*

P(IS) = 3/5

P(L|IS) = 2/3

P(-I|IS) = 0 … but 0.01 because of Laplace Smoothing

P(-S|IS) = 1/3

P(D|IS) = 1/3

P(A|IS) = 1/3

P(IS | L, -I, -S, D, A) = 3/5*2/3*0.01*1/3*1/3*1/3 = 0.000148

$P(DS \mid L, -I, -S, D, A) = P(DS) * P(L|DS) * P(-I|DS) * P(-S|DS) * P(D|DS) * P(A|DS)$

	L	I	S	D	A	Course
D1	T	T	T	F	F	IS
D2	F	T	F	T	T	DS
D3	T	T	F	F	T	IS
D4	F	T	T	T	F	IS
D5	F	F	T	T	T	DS
D6	*T*	*F*	*F*	*T*	*T*	*?*

$P(DS) = 2/5$

$P(L|DS) = 0.01$ (Laplace Smoothing)

$P(-I|DS) = 1/2$

$P(-S|DS) = 1/2$

$P(D|DS) = 1$

$P(A|DS) = 1$

$P(DS \mid L, -I, -S, D, A) = 2/5*0.01*1/2*1/2*1*1 = 0.001$

$P(DS \mid L, -I, -S, D, A) > P(IS \mid L, -I, -S, D, A)$

Then D6 would be categorized as DS.

You are given the following text documents (one sentence is a document for this exercise).

1. A cat sleeps near the field. label: passive
2. The cow walks in the field. label: active
3. The field is full of poppies. label: passive
4. The cats and the dogs run in the field. label: active
5. The cow is mad because the cats and the dogs walked all over the poppies. label:active

Q1: convert each of these sentences into feature vectors. Use the following preprocessing steps:

- Lowercase text
- Remove punctuation
- Stemming (you can do this intuitively)
- Remove stop words (remove these words: a, the, of, in, all, because, over, on)
- you do not need to count occurrences as all of them should be one in this example (no need to do tf-idf, but you can if you want to, but only do that if you are solving this with a computer program and not on the paper)

Tip: create a table to write these vectors down, like this fictive example for the words {table, chair, desk, lamp}.

Table	**Chair**	**desk**	**lamp**
1	1	0	0
0	1	0	1

Step 1 : Using the preprocessing steps, we are left with the following sets

1. {cat, sleep, near, field}
2. {cow, walk, field}
3. {field, is, full, poppy}
4. {cat, and, dog, run, field}
5. {cow, is, mad, cat, and, dog, walk, poppy}
6. Step 2 : convert into vectors, each unique word is a dimension

cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1	1	1	0	0	0
1	0	0	1	0	0	1	0	0	0	1	1	0
1	0	0	0	1	1	1	1	0	1	1	0	1

Exercise 7: K-NN for text classification

E7a) Now, you are creating a k-NN classifier (for the dataset from Exercise 6), with k = 2, and using Euclidean distance as a metric. To which class (passive or active) do these two sentences belong?

A. The dog walks in the field.
B. The cow sleeps on the poppies.

cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1	1	1	0	0	0
1	0	0	1	0	0	1	0	0	0	1	1	0
1	0	0	0	1	1	1	1	0	1	1	0	1

A. {dog, walk, field}
B. {cow, sleep, poppy}

cat	sleep	near	field	cow	walk	Dog	is	full	poppy	and	run	mad
0	0	0	1	0	1	1	0	0	0	0	0	0
0	1	0	0	1	0	0	0	0	1	0	0	0

DEFINITION

k-NN CLASSIFIER

Non-parametric method used for classification and regression that predicts the label of a data point by looking at the 'k' closest labeled data points and choosing the most common label among them

$$\sqrt{\sum_{i=1}^{13} (q_i - p_i)^2}$$

	cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	1	1	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	1	1	1	0	0	0
4	1	0	0	1	0	0	1	0	0	0	1	1	0
5	1	0	0	0	1	1	1	1	0	1	1	0	1
A	0	0	0	1	0	1	1	0	0	0	0	0	0
B	0	1	0	0	1	0	0	0	0	1	0	0	0

	1	2	3	4	5
A					
B					

	cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	1	1	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	1	1	1	0	0	0
4	1	0	0	1	0	0	1	0	0	0	1	1	0
5	1	0	0	0	1	1	1	1	0	1	1	0	1
A	0	0	0	1	0	1	1	0	0	0	0	0	0
B	0	1	0	0	1	0	0	0	0	1	0	0	0

	1	2	3	4	5
A	5				
B					

	cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	1	1	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	1	1	1	0	0	0
4	1	0	0	1	0	0	1	0	0	0	1	1	0
5	1	0	0	0	1	1	1	1	0	1	1	0	1
A	0	0	0	1	0	1	1	0	0	0	0	0	0
B	0	1	0	0	1	0	0	0	0	1	0	0	0

	1	2	3	4	5
A	5				
B					

	cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	1	1	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	1	1	1	0	0	0
4	1	0	0	1	0	0	1	0	0	0	1	1	0
5	1	0	0	0	1	1	1	1	0	1	1	0	1
A	0	0	0	1	0	1	1	0	0	0	0	0	0
B	0	1	0	0	1	0	0	0	0	1	0	0	0

	1	2	3	4	5
A	5	2			
B					

	cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	1	1	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	1	1	1	0	0	0
4	1	0	0	1	0	0	1	0	0	0	1	1	0
5	1	0	0	0	1	1	1	1	0	1	1	0	1
A	0	0	0	1	0	1	1	0	0	0	0	0	0
B	0	1	0	0	1	0	0	0	0	1	0	0	0

	1	2	3	4	5
A	5	2			
B					

	cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	1	1	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	1	1	1	0	0	0
4	1	0	0	1	0	0	1	0	0	0	1	1	0
5	1	0	0	0	1	1	1	1	0	1	1	0	1
A	0	0	0	1	0	1	1	0	0	0	0	0	0
B	0	1	0	0	1	0	0	0	0	1	0	0	0

	1	2	3	4	5
A	5	2	5		
B					

	cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	1	1	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	1	1	1	0	0	0
4	1	0	0	1	0	0	1	0	0	0	1	1	0
5	1	0	0	0	1	1	1	1	0	1	1	0	1
A	0	0	0	1	0	1	1	0	0	0	0	0	0
B	0	1	0	0	1	0	0	0	0	1	0	0	0

	1	2	3	4	5
A	5	2	5		
B					

	cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	1	1	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	1	1	1	0	0	0
4	1	0	0	1	0	0	1	0	0	0	1	1	0
5	1	0	0	0	1	1	1	1	0	1	1	0	1
A	0	0	0	1	0	1	1	0	0	0	0	0	0
B	0	1	0	0	1	0	0	0	0	1	0	0	0

	1	2	3	4	5
A	5	2	5	4	
B					

	cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	1	1	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	1	1	1	0	0	0
4	1	0	0	1	0	0	1	0	0	0	1	1	0
5	1	0	0	0	1	1	1	1	0	1	1	0	1
A	0	0	0	1	0	1	1	0	0	0	0	0	0
B	0	1	0	0	1	0	0	0	0	1	0	0	0

	1	2	3	4	5
A	5	2	5	4	
B					

	cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	1	1	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	1	1	1	0	0	0
4	1	0	0	1	0	0	1	0	0	0	1	1	0
5	1	0	0	0	1	1	1	1	0	1	1	0	1
A	0	0	0	1	0	1	1	0	0	0	0	0	0
B	0	1	0	0	1	0	0	0	0	1	0	0	0

	1	2	3	4	5
A	5	2	5	4	7
B					

	cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	1	1	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	1	1	1	0	0	0
4	1	0	0	1	0	0	1	0	0	0	1	1	0
5	1	0	0	0	1	1	1	1	0	1	1	0	1
A	0	0	0	1	0	1	1	0	0	0	0	0	0
B	0	1	0	0	1	0	0	0	0	1	0	0	0

	1	2	3	4	5
A	5	2	5	4	7
B	5	4	5	8	7

… and now, take the square root of the distances!

	cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	1	1	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	1	1	1	0	0	0
4	1	0	0	1	0	0	1	0	0	0	1	1	0
5	1	0	0	0	1	1	1	1	0	1	1	0	1
A	0	0	0	1	0	1	1	0	0	0	0	0	0
B	0	1	0	0	1	0	0	0	0	1	0	0	0

	1	2	3	4	5
A	2.24	1.41	2.24	2	2.65
B	2.24	2	2.24	2.83	2.65

	1	2	3	4	5
A	2.24	1.41	2.24	2	2.65
B	2.24	2	2.24	2.83	2.65

For vector A the nearest neighbours are: 2,4

1	2	3	4	5
Passive	Active	Passive	Active	Active

For vector A the nearest neighbours are: 2,4

1	2	3	4	5
Passive	Active	Passive	Active	Active

For vector A the nearest neighbours are: 2,4 =>
A is active

1	2	3	4	5
Passive	Active	Passive	Active	Active

For vector B the nearest neighbours are: 2, 1 or 3

We have 3 cases

1) Choose 2 and 1
2) Choose 2 and 3
3) Choose 2 and 1 and 3

1	2	3	4	5
Passive	Active	Passive	Active	Active

For vector B the nearest neighbours are: 2, 1 or 3

We have 3 cases

1) Choose 2 and 1
   - active/ passive => a tie
   - 2 is even closer => B is active OR
   - indecisive

1	2	3	4	5
Passive	Active	Passive	Active	Active

For vector B the nearest neighbours are: 2, 1 or 3

We have 3 cases

2) Choose 2 and 3

- active/ passive => a tie
  - 2 is even closer => B is active OR
  - indecisive

1	2	3	4	5
Passive	Active	Passive	Active	Active

For vector B the nearest neighbours are: 2, 1 or 3

We have 3 cases

3) Choose 2 and 1 and 3

- Active / Passive / Passive = > Passive

E7b): What would the result be if you use Hamming Distance and k3?

Hamming distance = #bits different / #bits

	cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	1	1	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	1	1	1	0	0	0
4	1	0	0	1	0	0	1	0	0	0	1	1	0
5	1	0	0	0	1	1	1	1	0	1	1	0	1
A	0	0	0	1	0	1	1	0	0	0	0	0	0
B	0	1	0	0	1	0	0	0	0	1	0	0	0

	1	2	3	4	5
A					
B					

	cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	1	1	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	1	1	1	0	0	0
4	1	0	0	1	0	0	1	0	0	0	1	1	0
5	1	0	0	0	1	1	1	1	0	1	1	0	1
A	0	0	0	1	0	1	1	0	0	0	0	0	0
B	0	1	0	0	1	0	0	0	0	1	0	0	0

	1	2	3	4	5
A	5/13				
B					

	cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	1	1	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	1	1	1	0	0	0
4	1	0	0	1	0	0	1	0	0	0	1	1	0
5	1	0	0	0	1	1	1	1	0	1	1	0	1
A	0	0	0	1	0	1	1	0	0	0	0	0	0
B	0	1	0	0	1	0	0	0	0	1	0	0	0

	1	2	3	4	5
A	5/13	2/13	5/13	4/13	7/13
B	5/13	4/13	5/13	8/13	7/13

1	2	3	4	5
Passive	Active	Passive	Active	Active

For vector A the best 3 training data are: 2, 4, 1 or 3

2 cases

1) 2 and 4 and 1 —---> active / active / passive => active
2) 2 and 4 and 3 —---> active / active / passive => active

1	2	3	4	5
Passive	Active	Passive	Active	Active

For vector B the best 3 training data are: 1, 2 ,3

Which give:
Active / Passive/ Passive => Passive

E7c): What would the result be if you use Cosine Similarity and k=1?

$$\frac{how\ many\ times\ both\ have\ 1}{\sqrt{number\ of\ 1\ in\ our\ knowledge\ base}} * \sqrt{number\ of\ 1\ in\ our\ example}$$

	cat	sleep	near	field	cow	walk	dog	is	full	poppy	and	run	mad
1	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	1	1	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	1	1	1	0	0	0
4	1	0	0	1	0	0	1	0	0	0	1	1	0
5	1	0	0	0	1	1	1	1	0	1	1	0	1
A	0	0	0	1	0	1	1	0	0	0	0	0	0
B	0	1	0	0	1	0	0	0	0	1	0	0	0

	1	2	3	4	5
A	$1/\sqrt{4}*\sqrt{3}$ = 0.28	$2/\sqrt{3}*\sqrt{3}$ = 0.66	$1/\sqrt{4}*\sqrt{3}$ =0.28	$2/\sqrt{3}*\sqrt{5}$ = 0.51	$2/\sqrt{8}*\sqrt{3}$ = 0.4
B	$1/\sqrt{4}*\sqrt{3}$ =0.28	$1/\sqrt{3}*\sqrt{3}$ =0.33	$1/\sqrt{4}*\sqrt{3}$ = 0.28	0	$2/\sqrt{8}*\sqrt{3}$ 0.4

1	2	3	4	5
Passive	Active	Passive	Active	Active

For vector A the best training data is: 1 or 3

Which give:
A => Passive

1	2	3	4	5
Passive	Active	Passive	Active	Active

For vector B the best training data is: 4

Which give:
B => Active